PBL-I Project Review – Machine Learning (21CSC305P)

Unit 1 & Unit 2 | 20 Marks

Group 4: Used Car Price Prediction and Condition Classification

Project problem: Develop an ML system to predict the resale price of used cars and classify vehicles according to their condition/value category.

Dataset: Group_4_Dataset.csv – 8,128 used-car listings, 13 raw columns (name, year, selling_price, km_driven, fuel, seller_type, transmission, owner, mileage, engine, max_power, torque, seats).

Q1. Problem Identification & Dataset Selection – 4 Marks

Real-world problem. Used-car marketplaces need to (a) quote a fair resale price for a listed vehicle and (b) bucket it into a value tier a buyer can filter by. An ML system can do both consistently from the vehicle's own attributes.

ML objective. *Regression:* predict selling_price (continuous, INR) from vehicle attributes. *Classification:* predict a price tier – Budget / Mid / Premium – a discretisation of the same target standing in for "condition/value category", since the raw data has no separate condition field. Tier cut points are the 33rd/67th percentile of price, fit on the training split only so no test information leaks into the label definition.

Input features (both tasks): car_age (derived from year), km_driven, mileage, engine, max_power, torque_nm, torque_rpm, seats (numeric) and brand (derived from name), fuel, seller_type, transmission, owner (categorical). selling_price is the regression target and is excluded from the classifier's features to avoid circularity.

Regression target: selling_price (modelled as log(selling_price)). **Classification target:** price_tier $\in$ {Budget, Mid, Premium}.

Dataset justification. 8,128 rows span 32 brands, model years 1983–2020, and all fuel/transmission/ownership categories with enough support per level for stratified learning (the rarest class, owner = Test Drive Car, still has 5 rows). It directly matches the brief's suggested data and is an appropriate size for the four regression and three classification models used in Q3.

Q2. Exploratory Data Analysis & Insights – 4 Marks

Cleaning pipeline (summary). Exact duplicates (1,202 of 8,128) are dropped first, before any split, so duplicates can never straddle train/test. Unit suffixes are stripped from mileage/engine/max_power; torque's many free-text formats are parsed into torque_nm and torque_rpm, correctly resolving an Nm-vs-kgm labelling ambiguity present in the source data. Undisclosed zero readings and a handful of physically-impossible torque values are treated as missing. Rows with no spec data at all (208) are dropped; the small residual NaN count is median-imputed. brand and car_age are engineered. **Final cleaned dataset: 6,718 rows, 0 missing values.**

Missing values (raw dataset)

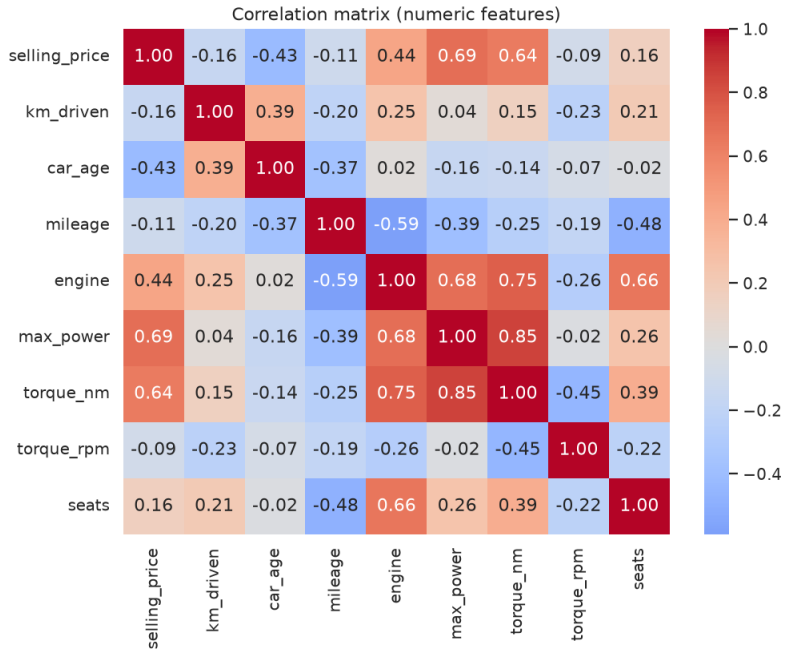
	missing_count	missing_pct
torque	222	2.73
seats	221	2.72
engine	221	2.72
mileage	221	2.72
max_power	215	2.65

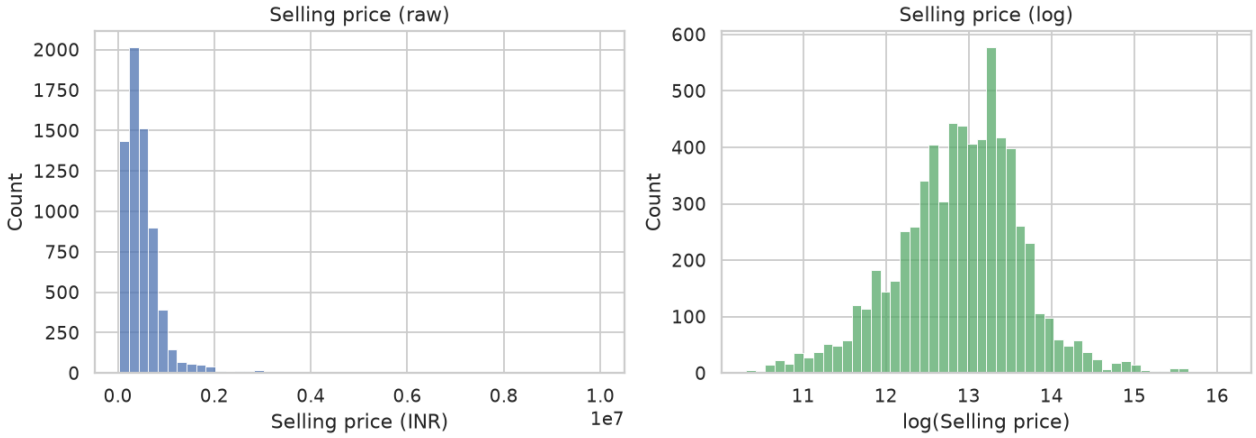
Descriptive statistics – mean, std, variance, quantiles (cleaned data)

	mean	std	var	min	q25	median	q75	q95	max
selling_price	5.26e+05	5.24e+05	2.74e+11	3e+04	2.5e+05	4.2e+05	6.5e+05	1.2e+06	1e+07
km_driven	7.34e+04	5.87e+04	3.45e+09	1	3.8e+04	6.82e+04	1e+05	1.55e+05	2.36e+06
car_age	6.39	3.9	15.2	0	3	6	9	13	26
mileage	19.5	3.94	15.5	9	16.8	19.5	22.5	25.8	42
engine	1.43e+03	493	2.44e+05	624	1.2e+03	1.25e+03	1.5e+03	2.5e+03	3.6e+03
max_power	87.8	31.7	1.01e+03	32.8	67.1	81.8	100	148	400
torque_nm	171	83.6	6.98e+03	47.1	110	160	200	330	640
torque_rpm	3.04e+03	889	7.9e+05	1.4e+03	2.25e+03	3e+03	4e+03	4.5e+03	5.3e+03
seats	5.43	0.984	0.969	2	5	5	5	7	14

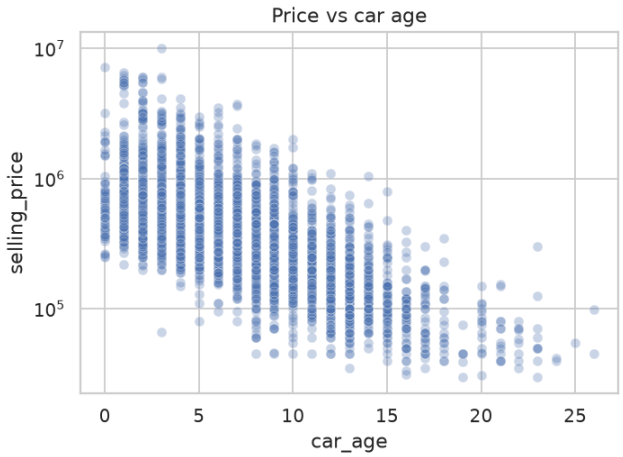
Correlation with selling_price

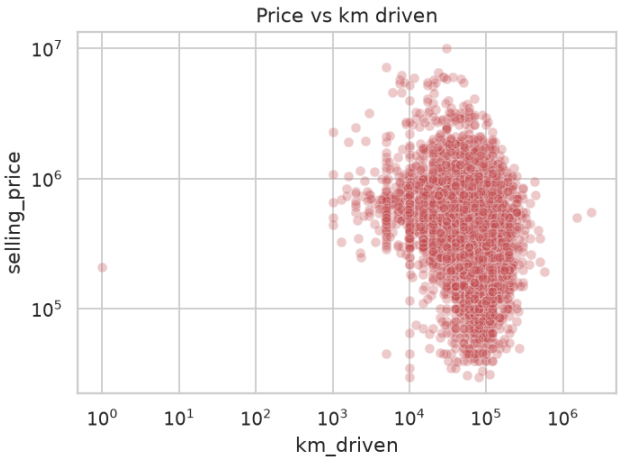
	correlation
selling_price	1
max_power	0.692
torque_nm	0.638
engine	0.443
seats	0.158
torque_rpm	-0.091
mileage	-0.108
km_driven	-0.161
car_age	-0.427

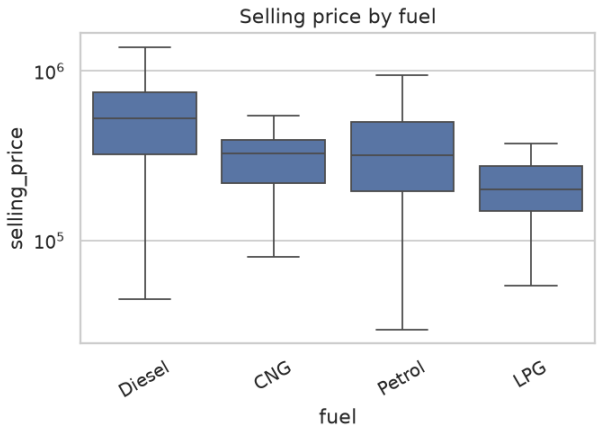


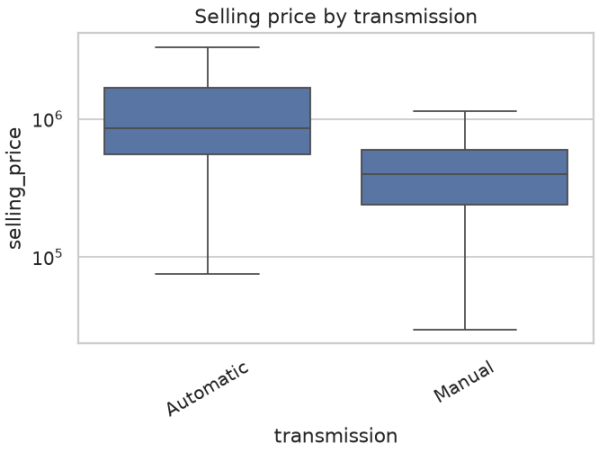


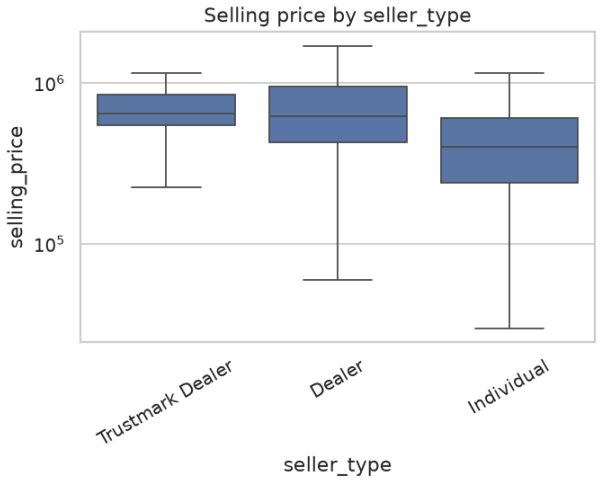
selling_price is strongly right-skewed; the log transform (right panel) is roughly symmetric, which is why regression targets log(selling_price).


Price vs car age
selling_price
car_age

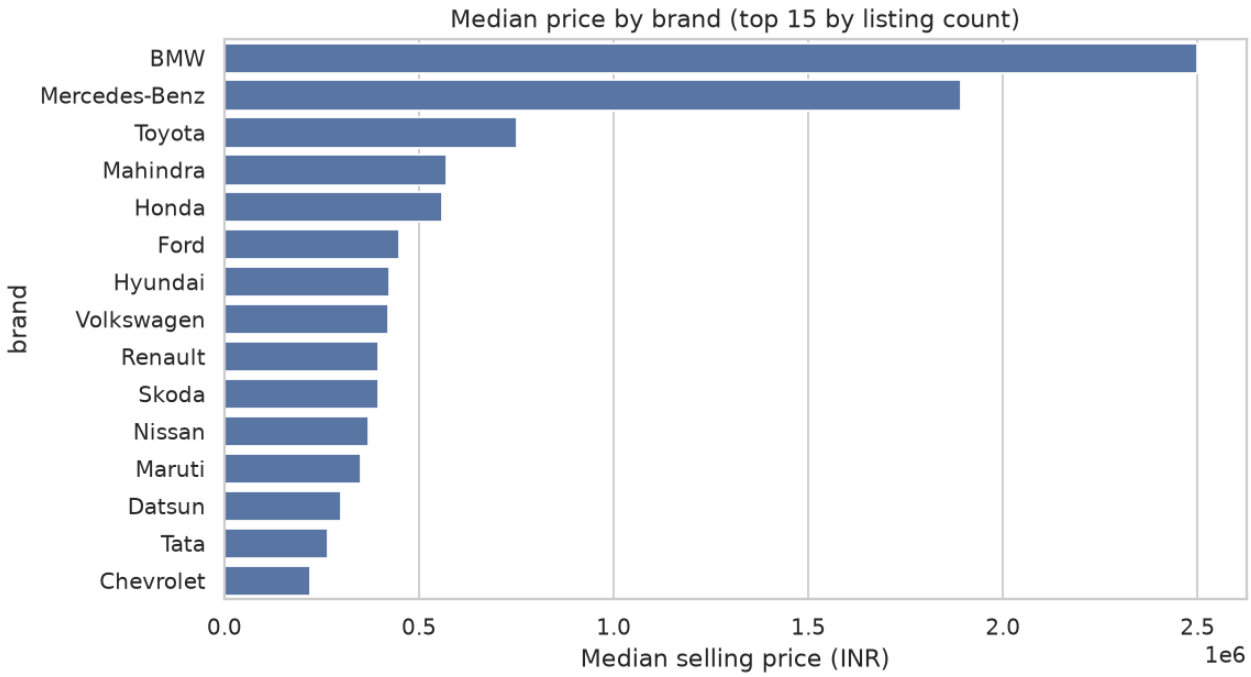

Price vs km driven
selling_price
km_driven


Selling price by fuel
selling_price
Diesel
CNG
Petrol
LPG
fuel


Selling price by transmission
selling_price
Automatic
Manual
transmission


Selling price by seller_type
selling_price
Trustmark Dealer
Dealer
Individual
seller_type


Selling price by owner
selling_price
Test Drive Car
First Owner
Second Owner
Third Owner
Fourth & Above Owner
owner


Median price by brand (top 15 by listing count)
BMW
Mercedes-Benz
Toyota
Mahindra
Honda
Ford
Hyundai
Volkswagen
Renault
Skoda
Nissan
Maruti
Datsun
Tata
Chevrolet
brand
0.0
0.5
1.0
1.5
2.0
2.5
1e6
Median selling price (INR)

Outlier summary (IQR rule)

	column	lower_fence	upper_fence	n_outliers	pct_outliers
0	selling_price	-3.5e+05	1.25e+06	306	4.55
1	km_driven	-5.5e+04	1.93e+05	161	2.4
2	car_age	-6	18	57	0.85
3	mileage	8.25	31.1	6	0.09
4	engine	746	1.95e+03	1207	18
5	max_power	17.8	149	304	4.53
6	torque_nm	-25.1	335	282	4.2
7	torque_rpm	-375	6.62e+03	0	0
8	seats	5	5	1464	21.8

seats and engine show the highest outlier rates by the strict 1.5×IQR rule, but both are genuinely multi-modal (5/7/8-seaters; small/mid/large engines) rather than data errors. selling_price and km_driven outliers are real high-end/high-mileage listings and are kept as-is (km_driven is capped only at the 99th percentile for modelling stability).

Key insights

• Price is right-skewed (mean INR 5.26L vs median INR 4.2L, max INR 1Cr) → modelled in log space for regression.

• car_age and km_driven are the strongest negative correlates of price (–0.43, –0.16); max_power and torque_nm are the strongest positive ones (+0.69, +0.64) – depreciation and engine output dominate resale value.

• Automatic-transmission and Diesel-fuel listings carry a materially higher median price than Manual/Petrol – a segment effect compounding a fuel-cost effect.

• max_power, engine and torque_nm are highly collinear ($r \approx 0.68$–0.85) – motivating Ridge/Bayesian regularisation over plain OLS in Q3.

Q3. Regression & Classification Implementation – 4 Marks

Preprocessing (shared). StandardScaler on numeric features, OneHotEncoder(drop='first', handle_unknown='ignore') on categoricals, assembled in a ColumnTransformer fit on the training fold only. Polynomial Regression additionally expands only the numeric block to degree 2 before the categorical columns are concatenated. A single 80/20 split (random_state=42), stratified on a rough price bin, is reused identically for both tasks so Q4's comparison is apples-to-apples.

Algorithms used (per the PDF's Q3 model list): *Regression* – Linear Regression, Polynomial Regression (degree 2), Ridge Regression (RidgeCV selects α), Bayesian Ridge Regression. *Classification* – Logistic Regression (multinomial), Linear SVM, RBF-kernel SVM. Regression target is log(selling_price); every reported metric is converted back to rupees before scoring.

Regression results

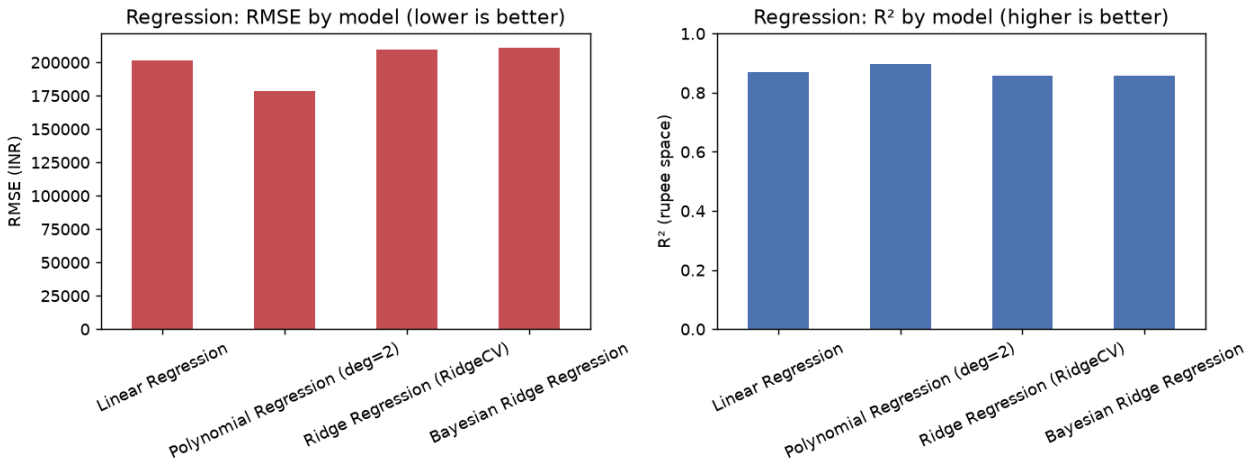
	MAE (INR)	MSE (INR^2)	RMSE (INR)	R2 (rupee space)	R2 (log space)
Linear Regression	9.53e+04	4.06e+10	2.01e+05	0.869	0.9
Polynomial Regression (deg=2)	8.81e+04	3.18e+10	1.78e+05	0.897	0.915
Ridge Regression (RidgeCV)	9.62e+04	4.4e+10	2.1e+05	0.858	0.9
Bayesian Ridge Regression	9.63e+04	4.45e+10	2.11e+05	0.856	0.899

Tier thresholds (from TRAIN prices only): Budget < INR 300,000 ≤ Mid < INR 555,000 ≤ Premium.

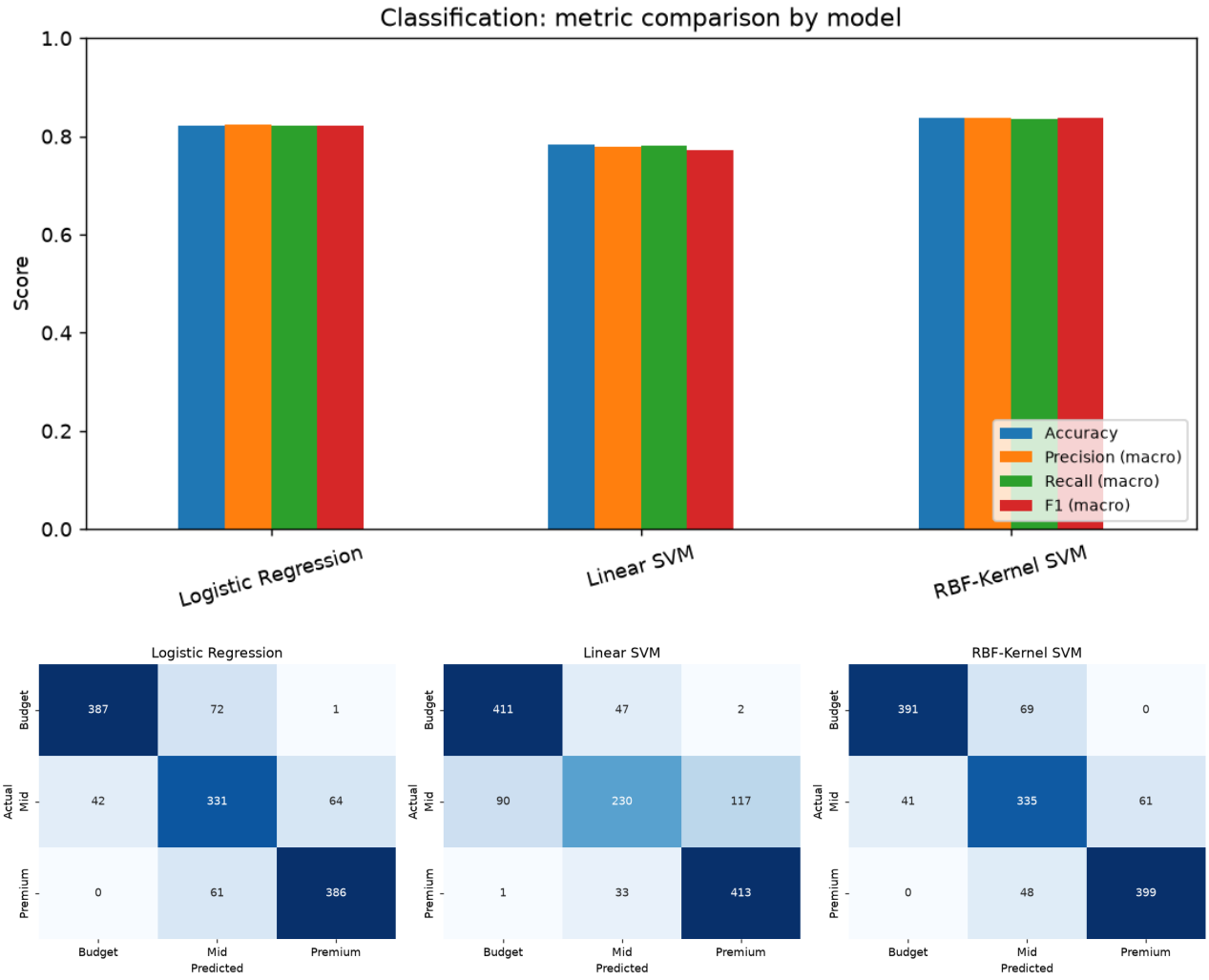
Classification results

	Accuracy	Precision (macro)	Recall (macro)	F1 (macro)
Logistic Regression	0.821	0.824	0.821	0.822
Linear SVM	0.784	0.779	0.781	0.771
RBF-Kernel SVM	0.837	0.838	0.836	0.837

Q4. Comparative Performance Analysis – 4 Marks



Polynomial Regression achieves the lowest RMSE and highest R^2 (≈0.90) – the numeric-feature interactions/curvature it adds are real and linear-only models can't capture them. Ridge and Bayesian Ridge score marginally below plain Linear Regression: with ~46 features on ~5,400 training rows the model isn't badly overfit to begin with, so regularisation mostly shrinks useful signal (RidgeCV itself selected a small α).



RBF-kernel SVM edges out Logistic Regression (≈0.84 vs ≈0.82 accuracy); the linear SVM trails (≈0.78). All three confusion matrices show the same pattern: confusion is concentrated almost entirely between

adjacent tiers, with near-zero Budget–Premium confusion – expected since the tiers are ordered price bands. The Mid class has the lowest per-class F1 for every model because it is bounded on both sides by a threshold and absorbs boundary error from both directions.

Cross-task consistency. The classifier is, by construction, a discretised version of the regressor over the same feature set and split – so their relative performance is expected to track rather than being two independent findings. Polynomial Regression's edge in R^2 and RBF-SVM's edge in accuracy both come from the same source: real non-linearity in how car_age/max_power/engine jointly determine price.

Q5. Result Interpretation & Presentation – 4 Marks

Top drivers of log(selling_price) – Linear Regression (standardized coefficients)

	feature	coefficient	abs_coefficient
0	owner_Test Drive Car	0.653	0.653
1	brand_Tata	-0.632	0.632
2	brand_Chevrolet	-0.572	0.572
3	brand_Land	0.522	0.522
4	brand_Datsun	-0.521	0.521
5	brand_Force	-0.51	0.51
6	brand_Ashok	-0.443	0.443
7	car_age	-0.435	0.435
8	brand_Fiat	-0.425	0.425
9	brand_Mahindra	-0.378	0.378



Coefficients are on standardized features, so they read directly as “effect per one std-dev change” and are comparable to each other. car_age is the single strongest, sign-consistent driver in both the regression and classification models – older cars are reliably worth less and more likely to fall in the Budget tier. brand dummies dominate the top-10 list for price, showing brand reputation/segment carries real, separable price information beyond the physical specs.

Practical significance

A ≈INR 88–95k MAE against a median price of INR 4.2L (≈2.7% of price; RMSE ≈ 4.2–5% of price) is workable for a guidance price – narrow enough for a listing platform to flag a quote as under/overpriced – but not tight enough to replace an inspector's appraisal for a single high-value transaction. The classifier's ≈82–84% accuracy suits filtering/browsing (a tier facet on a search page) where an occasional adjacent-tier misclassification is low-cost.

Limitations

• Single market snapshot (model years only run to 2020) – the model will drift as the market moves and would need periodic retraining.

• No accident/service history or physical inspection data – the classification task's “condition” is a value proxy (price tier), not true mechanical condition, which the raw dataset does not capture.

• mileage mixes two physical units (kmpl vs km/kg for CNG/LPG) merged into one numeric column; the fuel feature lets the models partially compensate.

• km_driven is capped at the 99th percentile for modelling, slightly under-weighting genuine but rare high-mileage listings.

Conclusion & future work

Both tasks confirm the same underlying structure: car age, engine output and brand jointly explain most of the price variation, with real non-linear interaction among them. Future work: fold in an external market-index feature to correct for the 2020 snapshot's staleness; test a spline/degree-3 basis on car_age specifically, since price-vs-age is visibly non-linear (steep early depreciation, flattening later); and, if genuine condition data becomes available, replace the price-tier proxy with a true condition label.